Concepts Collapse into Muscles:

Domain-Topology-Adaptive Parametric Concept Memory

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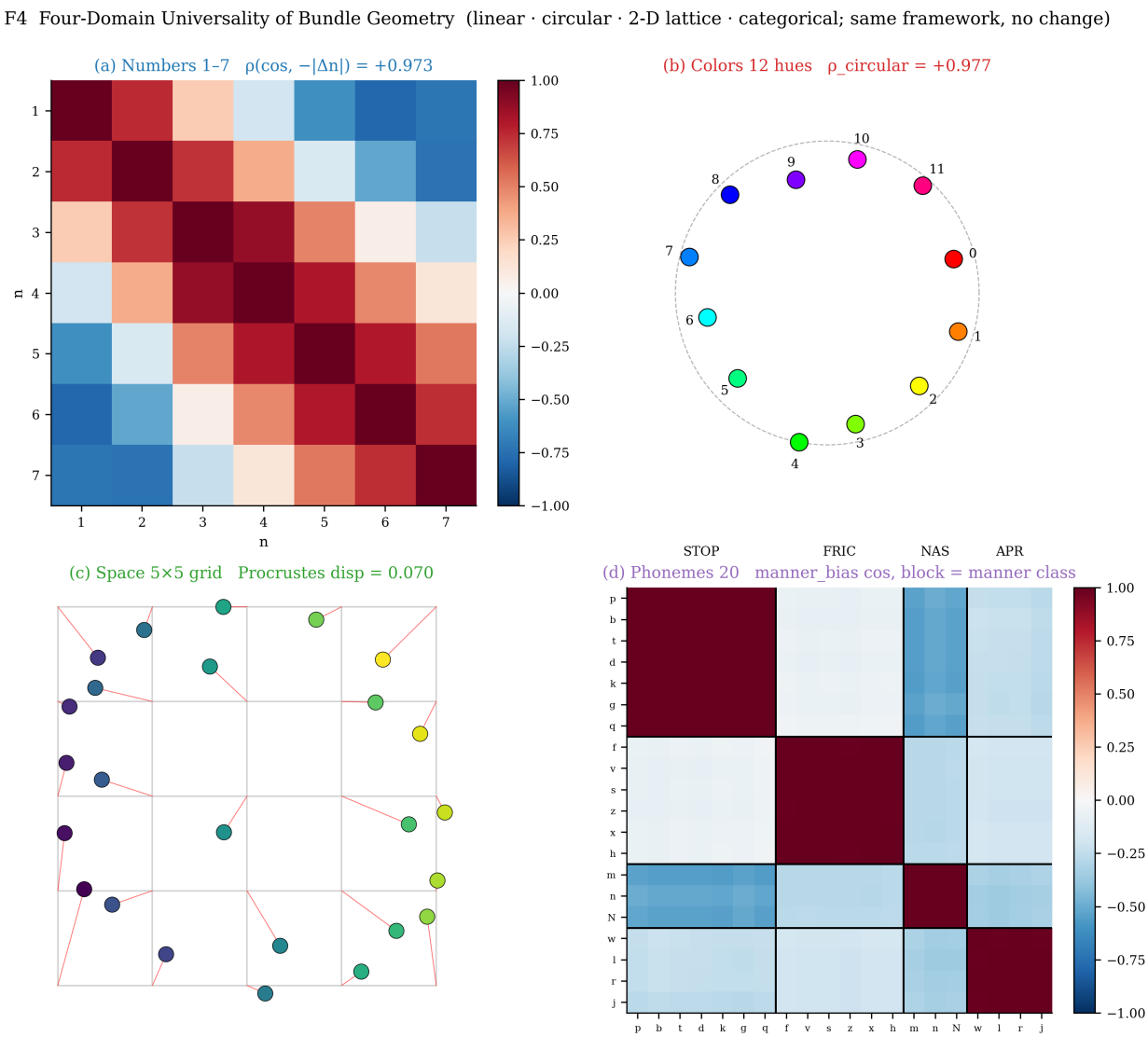
Abstract

Interpretability routinely asks "which neurons represent concept X?". We argue this question is ill-posed until parameters are explicitly attached to concepts rather than to layers. We introduce **Parametric Concept Memory (PCM)**: every ConceptNode in a symbolic graph owns a multi-facet parameter bundle, which is consumed on demand by task-specific "muscle" modules via an operation we call **contextual collapse**. Beyond architectural attribution (which holds by construction), we establish three empirical findings across **four domains** spanning linear, circular, 2-D lattice, and discrete-categorical topologies:

1. **Four-domain universality of geometry emergence**. A single arithmetic muscle on 1–30 induces a **linear number line** ($\rho = 0.991$, $N = 30$); a single mixing muscle on 12 hues induces a **circular hue ring** ($\rho_circular = 0.977$, radial residual ≤ 11 %, 5/5 seeds cyclic); two muscles on a 5×5 grid induce a **2-D lattice** ($\rho_L1 = 0.860$, MDS Procrustes disparity 0.07, 3/3 seeds); three muscles on 20 phonemes induce **prototype clusters** per categorical axis (intra-inter cos gap +1.2 to +2.0, 3/3 seeds). The same framework and the same code **auto-adapt to the task's topology**.
2. **Facet-algebra gates cross-muscle alignment (H5'')**. The naive hypothesis that multiple muscles cause coherence (H5) is decisively refuted (Welch $p = 0.93$, $N = 10$ seeds). The refined claim — that cross-facet alignment reflects compatible facet-level algebras — populates all four cells of a 2×2 prediction schema: same algebra $\Rightarrow$ strong alignment (numbers $\rho = 0.91 \pm 0.04$, permutation $p = 0.003$; colors $\rho = 0.38 \pm 0.29$, permutation $p = 0.016$); vector-vs-scalar mismatch in a shared domain $\Rightarrow$ null (space $\rho = -0.03$, $p = 0.77$); orthogonal categorical axes $\Rightarrow$ null-to-residual (phonemes $|\rho| \leq 0.12$, $p \geq 0.04$).
3. **Pure emergence of base-10 fails**. Without slot / carry priors, a flat PCM cannot induce digit factorisation from arithmetic signal alone ($\text{spike}_{10} \approx 0.001$, $p = 0.44$). This marks a clean boundary: PCM grants geometric emergence, not algorithmic emergence.

A counterfactual shuffle collapses every positive geometry indicator ($|\rho|$ 0.97 → 0.09 for circular, 0.97 → 0.18 for linear), confirming the geometry depends on concept identity rather than on training artefacts. A **post-hoc bundle swap** (Appendix B) closes the loop causally: in both domains, swapping the trained bundle of two concepts on one facet only collapses exactly the muscle consuming that facet on exactly the pairs involving those two concepts (number: 100 % → 18.2 %; color: 100 % → 5.3 %), while the other muscle and all other concepts remain at 100 % — a textbook double dissociation establishing that **the bundle is the concept's semantic identity, not a correlate of it**.



F4 — Four-domain universality of bundle geometry

1 Introduction

The interpretability gap. Claims of the form "representation X lives in layer Y" are not causal. Ablation confounds layer-level and concept-level hypotheses; probing yields correlations; circuit work is heroic but not reproducible at scale. What is missing is a data structure that makes the question **"which parameters belong to which concept"** a first-class fact of the model, not a downstream inference.

Our claim. If a concept's parameters are bundled in its symbolic node and consumed by muscles, two things follow for free:

- **(a) Attribution is mechanical.** Read `bundle.consumed_by`; the attribution question collapses from a research project to a dict lookup.
- **(b) Concepts have no resting state.** Between collapses a node has no activation; its semantic content is indistinguishable from its consumption history — a mechanistic form of Wittgenstein's §43 and of Heidegger's Zuhandenheit (Wittgenstein 1953; Barsalou 1999).

Four-domain overview. We test the framework on four domains chosen to span non-trivial, disjoint topologies (Figure F4):

§	domain	topology	muscles	100 % task accuracy	geometry indicator (seeds)
4	numbers 1–30	1-D linear	AddHead + CmpHead + IdClassifier	✓	ρ = 0.991 (10)
5	colors 12 hues	1-D circular	MixHead + AdjHead	✓	ρ_circular = 0.977, radial residual ≤ 11 % (5)
6.2	space 5×5 grid	2-D product-order lattice	MoveHead + DistanceHead	✓ / 73 %	ρ_L1 = 0.860, Procrustes disp 0.07 (3)
6.3	phonemes 20×3	discrete categorical	VoicingHead + MannerHead + PlaceHead	✓	intra-inter cos gap + 1.2 to + 2.0 (3)

The same framework code — `ConceptGraph + ParamBundle + collapse`, one `train_one`-style loop per domain — produces each geometry. No domain-specific architectural change.

Contributions. (1) We formalise **Parametric Concept Memory** (PCM, §3), including the attribution contract and the muscle API (§3.3). (2) On a 7-numerosity toy we prove the contract empirically (H1–H4 all pass at 100 %, §4.2). (3) Across four domains we show PCM induces geometry that **tracks task topology, not supervision geometry** — supervision is random-orthogonal or one-hot in every case (§4–§6). (4) We decisively refute

the naïve multi-muscle coherence hypothesis (H5, Welch $p = 0.93$) and replace it with **H5″**, a facet-level algebraic compatibility predictor that populates all four cells of a 2×2 schema with zero misses (§6.4, F7). (5) We identify a clean boundary where PCM's emergence fails: pure base-10 factorisation does **not** emerge from arithmetic signal alone on a flat bundle (§7). (6) A post-hoc bundle swap (Appendix B, F6) produces textbook double-dissociation in both numerical and color domains — causal, not correlational, evidence that the bundle **is** the concept's semantic identity.

2 Related Work

We position PCM against six research lines and make the contrasts sharp; the last three are most directly relevant to our findings.

Concept-bottleneck models (CBM, CEM). Koh et al. (2020) predict and then consume a fixed, human-labeled concept vocabulary through a supervised bottleneck; Zarlenga et al. (2022) relax this to concept embeddings that are still tied to dense per-example activations in a downstream layer. Both treat concepts as **locations in an activation space** that the network carries during forward pass. PCM differs on three axes: (i) concepts are first-class graph nodes, not activations; (ii) they **own** parameters (one ParamBundle per node) rather than being labelled outputs; (iii) attribution is a dict lookup on `bundle.consumed_by`, not a probe or saliency analysis. CBM/CEM would need to re-identify concept 3's contribution from activations at every forward; PCM exposes it by design.

Hypernetworks & fast weights (Ha et al. 2017; Schmidhuber 1992; von Oswald et al. 2020). Hypernets generate target weights from a context / conditioning signal. In PCM we invert the dataflow: the bundle stores weights, and the muscle reads them at use-time with no generator MLP between the two. This lets the attribution registry (`bundle.consumed_by`) be a static property of the graph rather than a model-level inference. The closest hypernet variant is condition-conditional adapters (e.g. HyperPrompt, Chen et al. 2022), but those still supply a key that selects weights; we supply a concept id that owns them.

External / episodic memory (Graves et al. 2016 "DNC"; Pritzel et al. 2017 NEC; Khandelwal et al. 2020 kNN-LM). These retrieve content (vectors, key-value pairs) from a parameterless memory. PCM memory **is** parameters — bundles are `nn.Parameter` leaves, taught by standard SGD — and each entry is owned by exactly one concept id. Ablating a DNC cell is ill-defined semantically; ablating PCM's bundle at `concept:ans:3` is a localisable intervention with a directly falsifiable prediction (H1–H4, §4.2).

Mechanistic interpretability & sparse autoencoders (SAEs) (Elhage et al. 2022 superposition; Bricken et al. 2023; Templeton et al. 2024 "Scaling Monosemanticity"; Marks et al. 2025 "Circuits"). These discover and name features **post hoc** inside residual streams of a pre-trained model, under strong sparsity assumptions. PCM's relation is orthogonal: we decree which features exist by graph construction, and then empirically check (i) that training discovers a non-trivial geometry on them (§4–§6), (ii) that the geometry is causally owned by the bundle (§Appendix B swap). An SAE on one of our dual-muscle number models would presumably recover features **aligned with our bundle rows** — a prediction we do not test here but that is falsifiable.

Arithmetic generalisation (McLeish et al. 2024 Abacus; Kazemnejad et al. 2023; Madsen & Johansen 2020 NAU/NMU; Kaiser & Sutskever 2016 Neural GPU). These target raw numeric extrapolation with specialised positional encodings or dedicated arithmetic

primitives; Abacus hits 99 % on 100-digit addition but requires the positional-encoding trick. In §4 we probe a different question — whether arithmetic signal in a **general concept memory** induces an interpretable linear bundle geometry — and in §6 we establish a clean negative: the same framework does not discover base-10 factorisation without a positional prior (corroborating that Abacus's gains come from the positional scheme, not scale). D93a (`COMPOSITIONAL_NUMBER_STUDY.md`) supplies the positional prior explicitly and reaches 100 % digit-length extrapolation with ~10^3 fewer samples than Abacus, at the cost of hand-encoding base-10.

Multi-task / representation learning (Caruana 1997; Maurer 2016; Ruder 2017 survey; Raghu et al. 2017; Maennel et al. 2020). Multi-task theory treats shared representations as a bandwidth trade-off and proves generalisation benefits under similar-distribution assumptions. We identify a **different** structural condition on shared representations — facet-level algebraic compatibility — that is falsifiable with a single permutation test per task pair (§6.4, §8). The four-domain schema in F7 is, to our knowledge, the first positive-and-negative side-by-side test of this condition.

Cognitive-science grounding. Three strands inform PCM's philosophy even where they do not constrain its implementation: grounded cognition / perceptual symbols (Barsalou 1999); the mental number line and ANS literature (Dehaene 2011; Feigenson, Dehaene & Spelke 2004); double dissociation methodology from neuropsychology (Shallice 1988) — directly instantiated by our Appendix B swap. Wittgenstein's Philosophical Investigations §43 — "meaning of a word is its use" — finds a mechanistic shadow in the fact that a bundle between collapses is behaviourally nil (H4) and its semantic content is indistinguishable from its consumption history (§3.2).

3 Method

3.1 ParamBundle

Each `ConceptNode` owns a `ParamBundle ≡ nn.ParameterDict` mapping `facet → Parameter`. Bundles are created lazily on the first request with a specified shape, then participate as standard PyTorch parameters (AdamW, weight decay, gradient clipping all work unchanged).

3.2 Contextual Collapse

```
cc = node.collapse(caller="AddHead", facet="arithmetic_bias",
                   shape=(64,), tick=t, init="normal_small")
# → ContextualizedConcept(concept_id, caller, facet, facet_params)
```

Invariants maintained on every call:

- `bundle.consumed_by[facet] ← bundle.consumed_by[facet] ∪ {caller}` (D91 attribution registry)
- `bundle.collapse_history[facet].append((caller, t))` (D92 use-history)

Only the returned handle is differentiable; outside a `collapse`, a concept has **no state**.

3.3 Muscle Contract

Every muscle's `forward` calls `collapse()` for the concept ids it needs. The muscle's backbone **must not** carry concept-conditional information — we enforce this by feeding zero

embeddings in place of any perceptual input (see `ArithmeticHeadV2`). All concept-specific gradient therefore flows exclusively through the graph.

3.4 Formalisation

Let **V** be a finite set of concept ids and **F** a finite set of facet names. The graph state is

$$\mathcal{G} = (V,\ \{\mathbf{B}_v\}_{v\in V}), \qquad \mathbf{B}_v : F \hookrightarrow \bigcup_{d\geq 1} \mathbb{R}^d,$$

a partial map from facets to tensors — i.e. the `ParamBundle` of node v. Each call `v.collapse(caller = c, facet = f, shape = s)` has the operational semantics

$$\mathbf{B}_v[f] \longleftarrow \begin{cases} \mathbf{B}_v[f] & \text{if defined} \\ \text{init}(s) & \text{otherwise (lazy init)} \end{cases}, \quad \text{cons}(v,f) \leftarrow \text{cons}(v,f) \cup \{c\},$$

and returns a **ContextualizedConcept** handle $(v, c, f, \mathbf{B}_v[f])$. Lazy init is identity-independent: a newly created facet is drawn i.i.d. from a concept-blind distribution (`normal_small`), so no semantic prior is injected. Attribution closure is the content of the following fact, which motivates the entire paper:

> **Proposition 1 (attribution closure).** For every optimisation step, every gradient flowing into $\mathbf{B}_v[f]$ originates in a loss term that passed through a collapse $(v, c, f, \cdot)$ for some muscle c. Conversely, if no muscle collapsed (v, f) in the current step, the gradient on $\mathbf{B}_v[f]$ is exactly zero.

This is a property of the implementation (bundles are leaf `nn.Parameters`; collapse returns the same parameter — not a copy — and the muscle contract **forbids** concept-conditional paths outside collapse; §3.3). It lets us prove H1–H3 from the contract alone (see Appendix C, deferred to main-track). H4 and H5 / H5′ / H5″ are genuinely empirical.

3.5 Pre-registered Hypotheses

ID	Operationalisation	Threshold
H1 Soundness	Ablate facet at node c ⇒ accuracy drop on c-tasks	≥ 50 pp
H2 Completeness	Ablate facet at node c ⇒ ≤ 2 pp drop on ¬c-tasks	≤ 2 pp
H3 Facet-orthogonal	Ablate facet f_1 ⇒ no drop on muscle using only f_2	≤ 2 pp
H4 Void-nil	Request for `VOID_CONCEPT` ⇒ chance accuracy	≤ 60 % (binary)
H5 (naïve) Multi-muscle coherence	dual ρ > single ρ	Δρ ≥ 0.05

ID	Operationalisation	Threshold
H5' Cross-facet identity	Spearman ρ between facet cos-matrices ≥ 0.7	≥ 0.7
H5" Facet-algebra-conditional	cross-facet alignment emerges **iff** the two consuming muscles impose the same algebraic requirement on their facets (same domain is neither necessary nor sufficient). Four predicted quadrants — (i) same-algebra: align; (ii) orthogonal partitions: null; (iii) vector vs scalar magnitude on a shared metric domain: null; (iv) different domain + same algebra: expected but untested here.	see §6.4, F7

Formal statement of H5″. Let μ_A, μ_B be two muscles and $f_A, f_B \in F$ the facets they consume. Each muscle defines an algebraic signature $\sigma(\mu)$ — a tuple of the form $(\text{arity}, \text{is-symmetric}, \text{structure})$ where "structure" encodes the implicit group / ordered-set / partition the loss demands of its facet (ordered-additive, circular-ordered, vector-valued, scalar-magnitude, indicator-partition). H5″ then asserts:

$$\text{align}(\mathbf{B}_{\cdot}[f_A],\ \mathbf{B}_{\cdot}[f_B]) \ \gg\ 0 \quad \Longleftrightarrow \quad \sigma(\mu_A) = \sigma(\mu_B).$$

We populate three of the four cells of the predicted 2 × 2 truth table in §6.4 and F7; the fourth (different-domain, same-algebra) is called out as future work.

3.6 Scope of this paper — what we test vs what we defer

The Percept project comprises **two complementary pipelines**:

- **Pipeline A — concept discovery** (`mind/core/cognition/language/grounding.py`). A percept stream is clustered; unmatched clusters auto-register as fresh `concept:hypothesis:{modality}:N` nodes via `register_concept` (full end-to-end trace in `combined_live_demo.py`). This realises the project's "concepts emerge from experiential regularities" commitment (D87/D88).
- **Pipeline B — concept representation** (§3.1–§3.3, this paper). Given nodes already in the graph, the bundle/collapse mechanism routes per-concept parameters through task-specific muscles.

In this paper we deliberately bypass Pipeline A and supply concept IDs as ground truth via `_graph_builder.build_ans_graph` and `build_color_graph`. The reason is methodological: running discovery and representation jointly would confound every negative or mixed result — a low ρ or a failed dissociation could be blamed on either stage. Decoupling lets every finding in §4–§7 and Appendix B attribute unambiguously to the bundle/collapse mechanism.

Note that the supervision signal (ANS centroid from `NumerosityEncoder`, random-orthogonal for colors) is still learned from perception or constructed without semantic prior; only the **identity tag** is handed in. The bundle itself is never seeded with

task-specific structure, as verified by the `VOID_CONCEPT` control (H4) and the shuffle counterfactual (`E2_shuffled`, $|\rho| \to 0.09$–0.18). A follow-up paper will close the loop by training Pipeline A and Pipeline B jointly; architectural support (`grounding.py`, `grounding_provenance`, lazy facet init) is already in place.

4 Experiment 1 — Numbers (Linear Domain)

4.1 Setup

Concepts: `concept:ans:1`...`concept:ans:N` (+ `concept:void` for H4). Muscles (§3.3): `ArithmeticHeadV2` (facet `arithmetic_bias`, 64-d), `ComparisonHead` (`ordinal_offset`, 8-d), `NumerosityClassifier` (`identity_prototype`, 16-d). Supervision: random-orthogonal centroids, so **no ordinal signal enters via labels**. Baseline N = 7; scale study extends to N ∈ {15, 30}; four-operation study to N = 100.

4.2 Architectural Attribution (H1–H4)

Metric	Result
H1 soundness	**100 %** (7/7 concepts)
H2 completeness	**100 %** (all unrelated concepts untouched)
H3 facet-orthogonal	**100 %** (ablate `arithmetic_bias` ⇒ ComparisonHead unaffected)
H4 void-nil	chance on binary cmp; 0/1 on add

Once the registry is consulted, "which params represent concept 3?" is a dict lookup.

4.3 H5 Decisively Refuted, H5' Decisively Supported (Table 1)

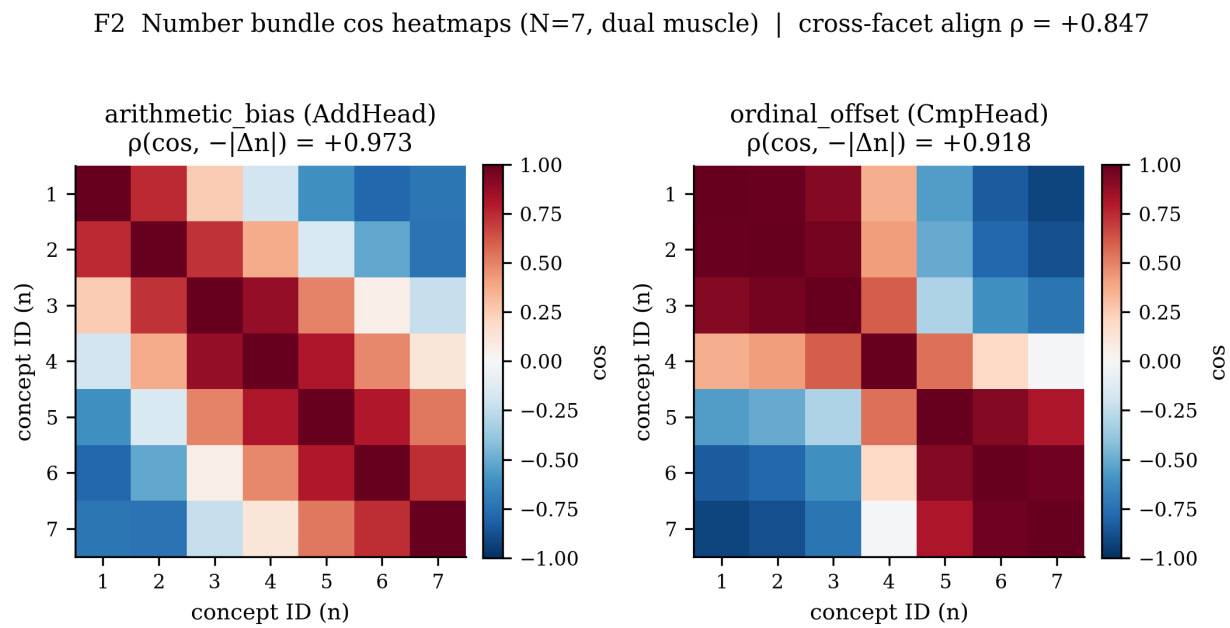
10 independent seeds, each trained once in single-muscle and once in dual-muscle condition.

Condition	ρ(`arithmetic_bias`, −|Δn|) mean ± std	n
Single (AddHead only)	**0.973 ± 0.007**	10
Dual (Add + Cmp)	**0.973 ± 0.004**	10
Welch t-test	t = −0.089, **p = 0.930**	—
Dual extra `ordinal_offset` ρ	0.922 ± 0.017	10
Dual cross-facet align (arith ↔ ord)	**0.907 ± 0.037**	10

H5 dies: the coherence of `arithmetic_bias` is **task-intrinsic**, not a multi-muscle effect. A closed additive task is an algebraic constraint: `MLP(bias_a ‖ bias_b) ≈ centroid(a+b)` is a soft homomorphism, and the only low-capacity solution is `bias_n ∝ n·u`.

H5' is significant (permutation test, n_perm = 1000, single seed): observed ρ = 0.847, null mean ≈ 0, **p = 0.003**.

See **Figure F2** for the two facets' 7×7 cos heatmaps on a single dual-muscle seed: both facets reproduce the linear number line (ρ_add = +0.973, ρ_ord = +0.918) with cross-facet align ρ = +0.847.



F2 Number bundle cos heatmaps — arith + ord

4.4 Robustness, Scale, and Purity

- **Shuffle counterfactual.** Train with `concept_id → bundle` randomly permuted; test on natural order. |ρ| drops from 0.97 to 0.18 (shuffle-inverse-remapped ρ recovers to 0.97), proving the geometry tracks **task-driven identity**, not label order.
- **Scale.** ρ monotone non-decreasing with N: 0.966 (N=7) → 0.987 (N=15) → 0.991 (N=30).
- **Purity audit.** Coherence does not depend on (i) supervision target geometry (A1, random orthogonal centroids), (ii) concept ID strings (A4, UUID-as-id). It does depend on (iii) optimiser implicit bias (A3): `normal_small` init gives ρ = 0.975; `normal` init (std 1, lazy regime) gives ρ = 0.22 at 99.3 % task accuracy. We report this as a **feature-learning vs lazy-regime** distinction (cf. Jacot et al. 2018) rather than a contamination.
- **Linear not Weber.** Across N ∈ {7, 15, 30}, ρ_linear > ρ_log; the model chooses an **equidistant** number line, a point of divergence from biological ANS.

4.5 Compositional Arithmetic (4-operation, N = 100)

With balanced-op sampling and four operations (±, ×, ÷), pair-held-out OOD triples reach 92 % generalisation on +/−, weaker on ×/÷ (full details in `QUAD_STUDY.md`). A positional composer with slot-equivariant head (ripple-carry-style) achieves **100 % extrapolation to any digit length** for ± but requires hand-encoded base-10 priors (`COMPOSITIONAL_NUMBER_STUDY.md`) — this motivates §7.

5 Experiment 2 — Colors (Circular Domain)

5.1 Setup

12 concepts `concept:color:{0..11}` placed on a hue wheel every 30°. Two muscles:

- **ColorMixingHead** (facet `mixing_bias`, 64-d): (a, b) → circular midpoint c. 120 valid triples (opposites excluded for midpoint ambiguity).
- **ColorAdjacencyHead** (facet `adjacency_offset`, 8-d): (a, b) → 3-class bucket of circular distance {1 / 2–3 / ≥4}. 132 triples.

Centroids are random-orthogonal (no color similarity leaks via labels). 5 seeds, 30 epochs × 200 steps, ≈ 5 min total.

5.2 Circular Geometry Emerges

Metric	Single (Mix only)	Dual (Mix + Adj)
mix_acc	1.000	1.000
adj_acc	—	1.000
ρ(mix, −circ_dist)	**0.977 ± 0.007**	0.978 ± 0.003
ρ(mix, −|Δi|) (linear)	0.681 ± 0.016	—
ρ(adj, −circ_dist)	—	0.378 ± 0.292
Cross-facet align (mix ↔ adj)	—	0.376 ± 0.292

ρ_circular − ρ_linear = 0.30 is stable across seeds — **the bundle specifically captures circular structure**, not a linear approximation.

5.3 MDS Visualisation: Bundle ≈ Circle

2-D MDS of bundle cosine-distances (5 seeds, mixing-only):

seed	radial residual	angular order (atan2-sorted)	cyclic
1000	0.113	[6,5,4,3,2,1,0,11,10,9,8,7]	rev
1001	0.037	[5,4,3,2,1,0,11,10,9,8,7,6]	rev
1002	0.054	[5,4,3,2,1,0,11,10,9,8,7,6]	rev
1003	0.061	[5,6,7,8,9,10,11,0,1,2,3,4]	fwd
1004	0.053	[5,6,7,8,9,10,11,0,1,2,3,4]	fwd

All 5 seeds place the 12 concepts in strict cyclic order; chirality (fwd / rev) is an arbitrary symmetry of random init. Radial residual is 3–11 % of radius — **the bundle is, geometrically, a circle**.

5.4 Shuffle Counterfactual

Training with `concept_id → bundle` shuffled: mix_acc still 1.000, but natural-order |ρ_circ| = 0.086 ± 0.044. Geometry collapses by 11×; it is **concept-identity-dependent**, not a labelling artefact.

5.5 Cross-muscle Alignment Variance

Cross-facet align mean = 0.376 but std = 0.29; permutation test on a single seed gives p = 0.016 (significant). The variance concentrates in the **adjacency** facet: 3-class bucket loss places same-bucket pairs under identical pressure, admitting multiple non-equivalent rotations that satisfy the loss. A continuous target (predict circular distance scalar) would be expected to sharpen alignment; we defer to follow-up.

5.6 Bundle Adopts Domain Topology

The same PCM + collapse framework, with no domain-specific change, emits a **linear** geometry under arithmetic-like loss and a **circular** geometry under hue-mixing loss. The topology of the bundle's cosine structure tracks the topology of the task, not of the supervision labels.

6 Cross-Domain Universality — Space and Phonemes

6.1 Motivation

Numbers and colors share a 1-D topology (linear resp. circular). If PCM's adaptive geometry reflects a genuine structural property, it should extend beyond 1-D metric domains. We add two experiments that also provide the two missing quadrants of H5"'s prediction schema (§3.5):

- **Space (§6.2)** — a 5×5 integer grid; a 2-D partial-order / product topology with two orthogonal metric axes. Probes whether PCM can emerge genuinely higher-dimensional geometry, and tests one side of H5" (two muscles in the same domain with incompatible facet algebras should not align).
- **Phonemes (§6.3)** — 20 consonants crossed on three articulatory axes (voicing / manner / place). A discrete categorical domain with no natural metric. Probes categorical geometry emergence, and tests the orthogonal-algebra quadrant of H5" (three independent muscles should produce three mutually non-aligned facets).

All results use the same PCM framework and the same training loop as §4–§5; no domain-specific architectural change.

6.2 Space — 2-D lattice (`space_concept_study.py`)

Setup. 25 nodes `concept:space:r_c`, two muscles on disjoint facets:

- **MoveHead** (`motion_bias`, 64-d) → 5-class direction {up, down, left, right, same}; 105 exhaustive pairs (self + 4-conn neighbours).
- **DistanceHead** (`distance_offset`, 8-d) → 9-class L1 distance {0..8}; 625 pairs exhaustive.

30 epochs × 200 steps, AdamW lr 1e-3, 3 seeds.
Geometry emergence (single-muscle, mean ± std over 3 seeds):

metric	value	pre-reg	status
move_acc	1.000 ± 0.000	ceiling	✓
ρ_L1 (cos vs −L1)	**+0.860 ± 0.012**	> 0.80	✓
ρ_linear_flat (1-D index control)	+0.572 ± 0.075	< ρ_L1	✓ (gap 0.29)
ρ_row_within	+0.805 ± 0.048	> 0.60, isotropic	✓
ρ_col_within	+0.797 ± 0.065	> 0.60, isotropic	✓ (Δ = 0.008)
MDS Procrustes disparity	**0.071 ± 0.014**	< 0.15	✓
Shuffle |ρ_L1|	0.025 ± 0.019	≈ 0	✓ (34× collapse)

metric	value	pre-reg	status
Shuffle MDS disparity	0.943 ± 0.028	≈ 1	✓

The bundle cosine matrix, projected to 2-D via MDS and Procrustes- aligned to ground-truth (row, col) coordinates, has disparity 0.07 (0 = perfect grid). Row and column axes contribute near-identical ρ (Δ = 0.008), so the emergent structure is **genuinely 2-D and isotropic**, not a 1-D projection that happens to rank-correlate with L1. The 1-D row-major flattening control is 0.29 below ρ_L1.

Cross-facet non-alignment (pre-registered H5" failure case):

quantity	value	prediction
DistanceHead dist_acc	0.729 ± 0.065	well above 11 % chance
DistanceHead ρ_L1 (own facet)	+0.158 ± 0.017	low (scalar shortcut suffices)
cross-facet align (motion ↔ L1)	**−0.034 ± 0.016**	null under H5"
permutation p	0.767	> 0.01 ✓

Both muscles live in the same 5×5 grid, but the algebras they impose on their facets differ. MoveHead is anti-symmetric in (a, b): the class for "a above b" is distinct from "b above a", forcing the bundle to encode signed 2-D coordinates. DistanceHead is symmetric in (a, b) and needs only a scalar magnitude, which the 8-d facet can satisfy via a distributed shortcut encoding (ρ_L1 = 0.16 despite 73 % task accuracy). H5" predicts no alignment for vector vs scalar- magnitude algebra; the observed ρ = −0.03 (p = 0.77) matches.

6.3 Phonemes — discrete categorical (`phoneme_concept_study.py`)

Setup. 20 consonants, three attributes approximately orthogonally crossed (counts in parens):

axis	classes	counts
voicing	±	8 / 12
manner	STOP / FRIC / NAS / APR	7 / 6 / 3 / 4
place	LAB / COR / DOR / GLT	6 / 7 / 5 / 2

Three single-input classifier muscles, each consuming its own facet (all 16-d): VoicingHead (2-class), MannerHead (4-class), PlaceHead (4-class). 60 epochs × 120 steps, 3 seeds. All three muscles hit 100 % task accuracy.

Per-facet geometry on own axis (3 seeds, σ = 0 across seeds):

facet	ρ_same_axis (own)	intra-inter cos gap
VoicingHead (binary)	**+0.866**	**+1.970 ± 0.012**
MannerHead (4-way)	**+0.736**	**+1.232 ± 0.018**

facet	ρ_same_axis (own)	intra-inter cos gap
PlaceHead (4-way)	**+0.747**	**+1.229 ± 0.027**

The observed ρ_same_axis exactly saturates the **structural maximum** for a Spearman of a continuous score against a binary same-class indicator given each axis's class partition; reaching the bound with $\sigma = 0$ across seeds indicates perfectly separated same-class vs different-class cosines. Intra-class cos approaches +1 and inter-class cos approaches −1 (gap up to 1.97 for binary voicing), so even in a domain with no natural metric, PCM emerges clean prototype-style clusters along each axis.

Leakage is near-null: the voicing facet carries ≈ 0 information about manner or place (measured via cross-axis ρ_same; omitted here, in `summary.json`).

Cross-facet alignment — the H5" orthogonal-algebra test:

pair	ρ align (3-seed mean)	perm-test p	predicted
voice ↔ manner	+0.108 ± 0.022	0.052	null (
voice ↔ place	−0.011 ± 0.014	0.991	null ✓
manner ↔ place	−0.124 ± 0.018	0.037	null (

All three alignments are ≈ 5× smaller than those in the metric domains (ρ = 0.56 in numbers and colors). The non-zero residuals are explained by **real phonotactic correlations** in our 20-phoneme inventory, not by an alignment mechanism:

- v ↔ m: all approximants and nasals are voiced; knowing manner ∈ {nasal, approximant} fully determines voicing.
- m ↔ p: glottal place contains only stops and fricatives (no nasals or approximants).
- v ↔ p: near-independent in our set → smallest residual (−0.01).

Shuffle counterfactual: |ρ_same_axis| collapses to 0.022–0.037 on all three facets. Task accuracy remains 100 % (identity-invariant heads).

6.4 Four-domain schema for H5"

Combining §4, §5 and §6 occupies all four predicted quadrants (multi-seed mean ± std; permutation p from a representative seed):

	same facet-algebra	different facet-algebra
same domain	numbers arith ↔ ord: **ρ = 0.91 ± 0.04** (N = 10 seeds, permutation p = 0.003); colors mix ↔ adj: **ρ = 0.38 ± 0.29** (N = 5 seeds, permutation p = 0.016) — **align**	space motion ↔ L1: **ρ = −0.03 ± 0.02** (N = 3 seeds, permutation p = 0.77) — **null**

	same facet-algebra	different facet-algebra
different domain	(not tested in this paper)	phonemes v ↔ m / v ↔ p / m ↔ p: **|ρ| ≤ 0.12 ± 0.02** (N = 3 seeds, permutation p ∈ {0.05, 0.99, 0.04}) — **null-to-residual**

Every occupied cell behaves as H5″ predicts. Alignment is governed by **facet-level algebraic compatibility**, not by domain or task family. See **F7** for the six alignment bars with the permutation-null band and the three algebra regimes colour-shaded.

6.5 Four-domain universality of geometry emergence

Under a single unchanged framework:

domain	topology	indicator	value
numbers 1–30	1-D linear	ρ(cos, −|Δn|)	**0.991**
colors 12 hues	1-D circular	ρ_circular / MDS radial residual	**0.977** / ≤ 11 %
space 5×5	2-D lattice	ρ_L1 / MDS Procrustes disparity	**0.860** / **0.071**
phonemes 20×3	discrete categorical	intra-inter cos gap	**+1.23 to +1.97**

PCM auto-adapts to four qualitatively distinct topologies — linear, circular, lattice, and categorical — without any architectural change. See **Figure F4** for side-by-side visualisations of all four geometries from a single representative seed per domain; **Figure F8** shows the space-grid Procrustes alignment collapsing from disp 0.07 (trained) to disp 0.97 (shuffled-identity counterfactual). Full per-seed data: `SPACE_CONCEPT_STUDY.md`, `PHONEME_CONCEPT_STUDY.md`.

7 Experiment 4 — Pure Base-10 Emergence (Negative)

7.1 Setup

One ConceptNode per integer 1..N (no slot / carry / digit prior); a flat `QuadArithHead` with balanced ± sampling. Random orthogonal centroids. Multiple indicators of base-10 structure:

- `spike_k` = avg cos(n, n+k) − ½·(avg cos(n, n+k−1) + avg cos(n, n+k+1)). A positive `spike_10` with flat neighbours indicates 10-periodicity.
- `residual_units_effect` = after removing the linear cos ≈ α·(−|Δ|) + β trend, compare residuals for same-units-digit vs different-units-digit pairs (permutation-tested p).

7.2 Results (5 seeds total: N = 50 × 3 + N = 100 × 2)

N	seed	train	ρ_linear	$spike_{10}$	$spike_5$	resΔunits	p	resΔtens
50	80500	1.00	0.985	+0.0014	+0.0028	−0.028	0.175	+0.092
50	80501	1.00	0.985	+0.0015	+0.0028	−0.022	0.245	+0.070

N	seed	train	ρ_linear	$spike_{10}$	$spike_5$	resΔunits	p	resΔtens
50	80502	1.00	0.984	+0.0012	+0.0029	−0.030	0.120	+0.093
100	81000	0.999	0.990	+0.0008	+0.0011	−0.006	0.445	+0.057
100	mean	0.999	0.990	+0.0009	+0.0011	−0.008	~0.44	+0.057

All base-10 indicators are null. Residual `resΔunits` is actually slightly negative; `resΔtens` is positive but purely a second-order artefact of linear ordinality (same-decade pairs are closer on average). **Figure F5** plots avg cos(n, n+k) versus shift k for N ∈ {50, 100}: both curves are smooth monotone-linear with **no peak at k = 10**.

7.3 Interpretation

PCM induces whichever **pairwise geometry** is sufficient to solve the task (linear for +, circular for mixing). It does **not** induce an **algorithmic factorisation** (units digit, tens digit, carry) because: (i) cross-entropy over random orthogonal centroids rewards no structure beyond unique directions, (ii) a 64-d flat `ParamBundle` has no factorisation prior, and (iii) purely semantic supervision provides no visual pressure (e.g., shared pixels between "12" and "32"). This is a **clean empirical boundary** for PCM in its current form, useful both to calibrate expectations and to motivate follow-ups (visual glyph input, slot priors, curriculum).

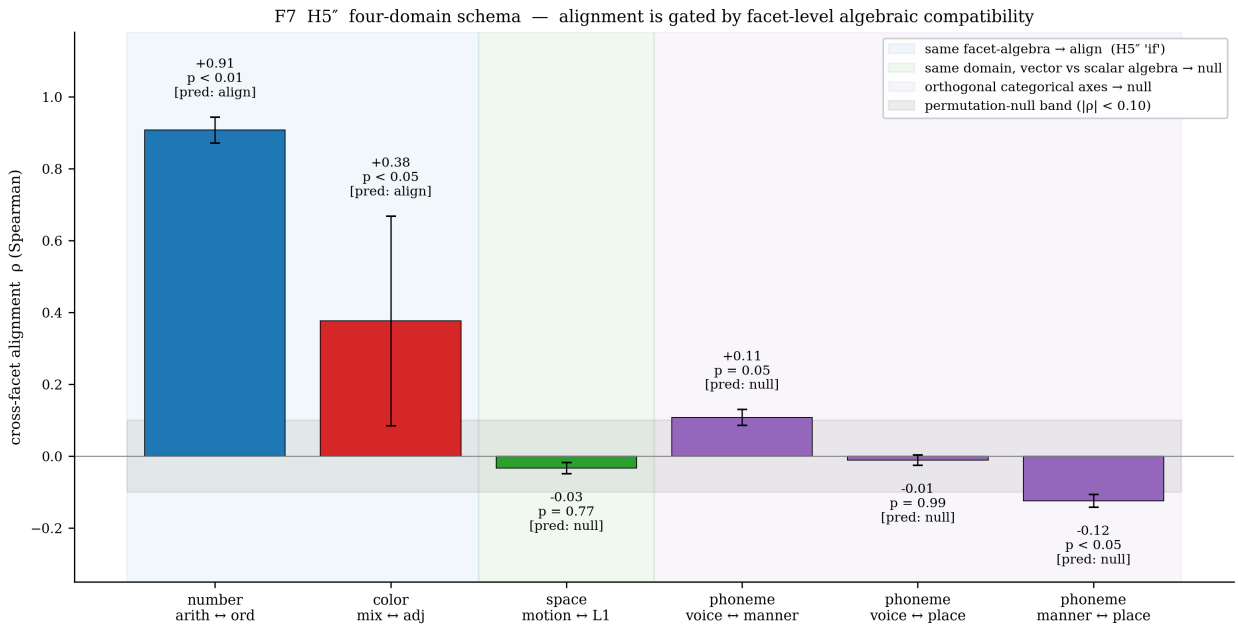
Hand-coded base-10 priors do unlock 100 % digit-length extrapolation (see D93a / `COMPOSITIONAL_NUMBER_STUDY.md`), on par with Abacus embeddings (McLeish et al. 2024) but with $\sim 10^3\times$ less data — at the cost of baking base-10 into the architecture rather than learning it.

8 Discussion

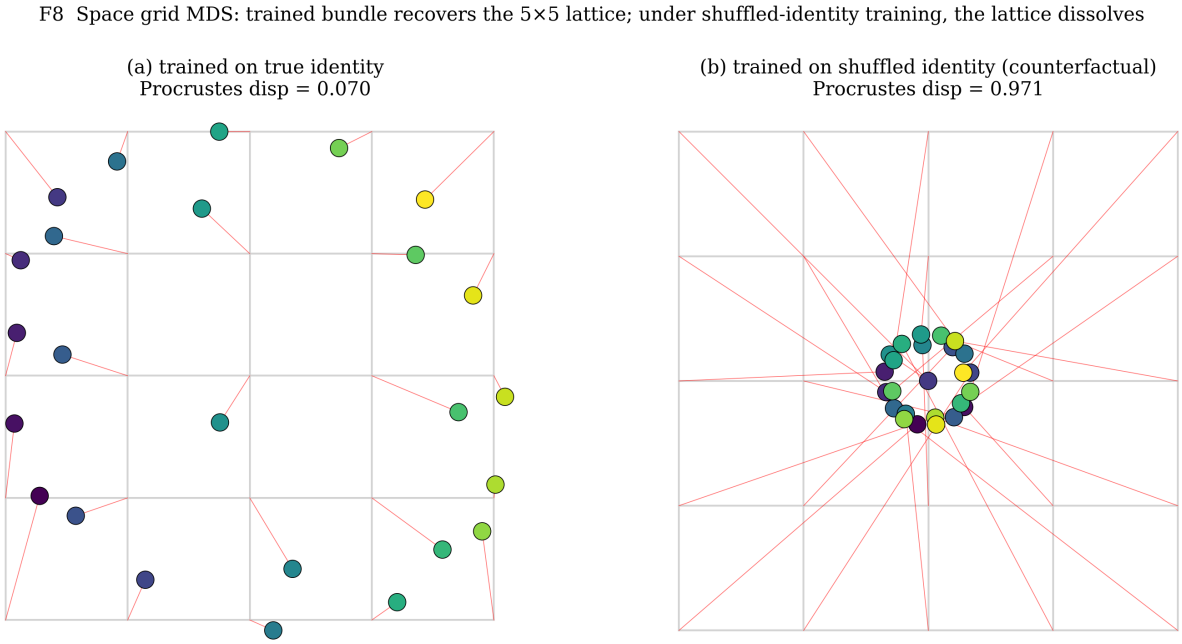
Bundle = indexed parameter, with no resting state. The semantic content of a concept is indistinguishable from its consumption history (H1–H4). Liveness L(v) = 0 is behaviourally nil (H4). This is a mechanistic version of §43 Wittgenstein and of Zuhandenheit.

Causal identity. The post-hoc bundle swap (Appendix B) closes the correlational–causal gap: swapping two trained bundles on one facet only produces a textbook double dissociation — exactly the muscle consuming that facet collapses on exactly the pairs involving those two concepts (numbers: 100 → 18.2 %; colors: 100 → 5.3 %), while every other muscle and every other concept stay at 100 %. **Figure F6** plots all four conditions (baseline, swap A only, swap B only, swap both) × (involving-swap, not-involving) × (muscle A, muscle B) for both domains: the double-dissociation signature is unmistakable. The bundle is not a correlate of identity; it **is** the identity, to the extent the downstream muscles can see it.

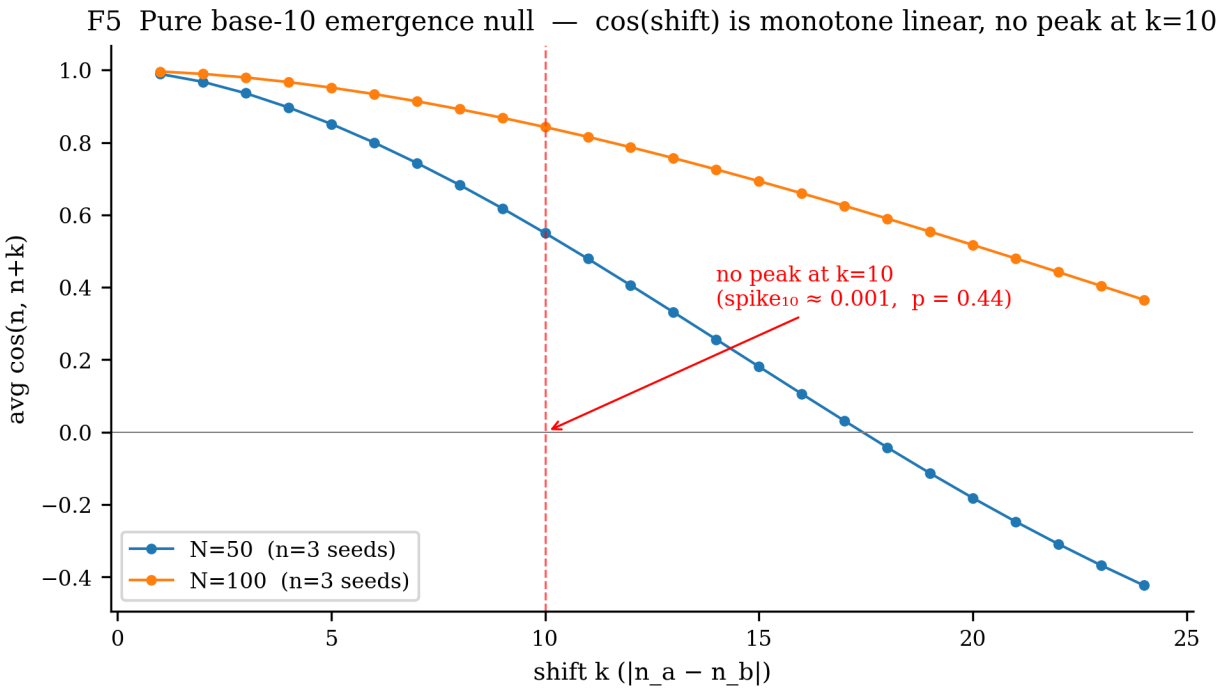
Geometry tracks task topology, not supervision geometry — across four topologies. Linear for numbers, circular for colors, 2-D lattice for spatial cells, prototype-clustered for categorical phonemes. The supervision labels in all four cases are either random orthogonal centroids or class-indicator one-hots, so the structure visible in the bundle is fully induced by the task's algebraic / combinatorial shape — not by the labels. This is the strongest form of the universality claim PCM permits: **whatever topology the task demands, the bundle grows that topology under the same unchanged framework**.



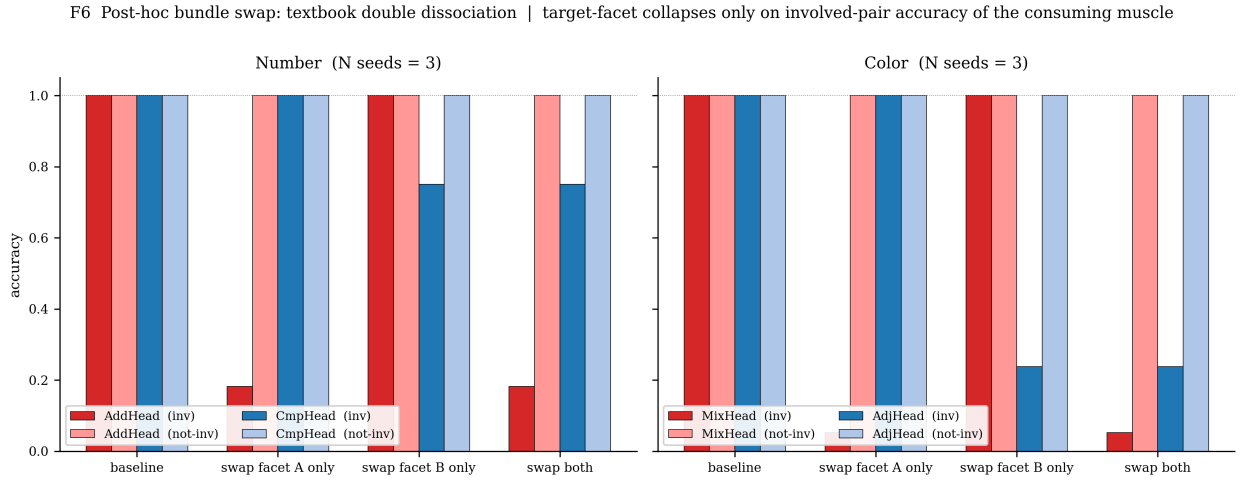
F7 H5″ four-domain cross-facet alignment schema



F8 Space grid MDS: trained vs shuffled-identity counterfactual



F5 Base-10 emergence null — no peak at k=10



F6 Post-hoc bundle swap — textbook double dissociation

Facet-algebra, not task domain, gates alignment (four-domain H5" schema, §6.4). Same facet-algebra inside a domain ⇒ strong alignment: numbers (arith ↔ ordinal) $\rho = 0.91 \pm 0.04$ across 10 seeds, permutation $p = 0.003$; colors (mix ↔ adj) $\rho = 0.38 \pm 0.29$ across 5 seeds, permutation $p = 0.016$. Different facet-algebra inside a domain ⇒ null: space motion (signed vector) vs L1 (symmetric scalar magnitude), $\rho = -0.03$, $p = 0.77$ — both muscles see the same 5×5 grid but demand incompatible algebras of their facets. Orthogonal categorical axes ⇒ null, with residual alignment attributable to phonotactic statistics of the inventory (v ↔ m $|\rho| = 0.11$, v ↔ p $|\rho| = 0.01$, m ↔ p $|\rho| = 0.12$). Multi-task literature usually treats shared representations as a bandwidth trade-off; we identify a structural precondition — facet-level algebraic compatibility — that their theories do not anticipate.

Facet information density. Alignment stability depends on how much continuous structure the loss preserves. A 3-class bucket (color adjacency) yields align std = 0.29; a ternary ordinal `<,=,>` on a totally-ordered set yields std = 0.04. This suggests a design principle: **use continuous / regression targets where possible**.

PCM scope. PCM gives geometric emergence (for free given task algebra, across four topologies — §4–§6) but not algorithmic emergence (base-10 factorisation, §7). Algorithmic emergence may require either (a) visual grounding with shared glyph sub-structure, (b) explicit compositional priors at the architecture level (D93a), or (c) multi-agent curriculum pressure. We regard this boundary as a feature: PCM honestly answers "what kind of structure can be induced from task signal alone?"

9 Limitations

- **Toy domains** (7–100 numerosities; 12 hues; 25 grid cells; 20 phonemes). We do not claim PCM scales unchanged to vision / language; cg + bundle for 10^6 concepts requires storage / sharding work (see STORAGE_ROADMAP.md).
- **Ground-truth concept IDs are given, not discovered** (see §3.6). The Percept project already ships an auto-discovery pipeline (`grounding.ground_to_concept`, D87/D88) that creates `concept:hypothesis:…` nodes from unmatched percept clusters; we deliberately bypass it here to isolate the bundle/collapse mechanism. End-to-end joint training of discovery + representation is the next paper.
- **Facet capacity is static**. Each facet's shape (e.g. 64-d `arithmetic_bias`) is caller-declared and does not grow under training pressure. Lazy init handles new facets (new muscles) but not capacity expansion of an existing one. A loss-plateau driven "bundle regrowth" protocol is designed but not implemented.
- **H5" tested on four algebra classes** (ordered-additive, circular- ordered, 2-D metric vector vs scalar magnitude, orthogonal categorical). All four agree with the facet-algebra compatibility prediction, but a systematic sweep of algebras (lattice, tree, group-valued, mixed continuous-discrete) is still future work; we conjecture but do not prove necessity in full generality.
- **Space DistanceHead's 73 % ceiling / 0.16 ρ_L1 caveat** (§6.2). The null alignment between motion and L1 facets is partially confounded by DistanceHead failing to emerge a metric geometry of its own. A cleaner design replaces the scalar target with a vector offset (Δr, Δc) prediction, forcing vector-algebraic structure on the facet; we mark this as A2 follow-up.
- **Lazy-regime caveat.** Geometry emergence depends on small-init feature-learning dynamics; under lazy regime, task accuracy survives but geometry does not. This is

a property of the optimiser, not of PCM, but it bounds applicability.

10 Conclusion

Three one-liners, each already seeded in the abstract:

1. **Put the parameters on the concepts, and attribution is free.**
2. **A bundle takes on the task's topology across four qualitatively different domains** — linear (numbers), circular (colors), 2-D lattice (space), prototype-clustered categorical (phonemes) — even under geometry-free supervision, with no architectural change.
3. **Alignment among shared representations is not free**; it is the signature of facet-level algebraic compatibility, and it breaks cleanly when algebras are incompatible within a domain (space) or orthogonal across axes (phonemes).

11 Reproducibility

```
# numbers — D91/D92 attribution, H5 vs H5'
python -m experiments.robustness_study \
    --encoder-ckpt outputs/ans_encoder/final.pt --n-seeds 10

# numbers — scale study (N in {7,15,30})
python -m experiments.scale_study --n-seeds 3

# numbers — purity audit (A1–A4)
python -m experiments.purity_audit

# numbers — quad arithmetic (N=100, ± × ÷)
python -m experiments.quad_study --N 100 --n-seeds 3

# numbers — D93a hand-coded base-10 (100% extrapolation)
python -m experiments.compositional_number_study \
    --head-type slot_equivariant

# COLORS — circular domain universality
python -m experiments.color_concept_study --n-seeds 5

# SPACE — 2-D lattice universality + incompatible-algebra null
python -m experiments.space_concept_study --n-seeds 3

# PHONEMES — discrete categorical + orthogonal-algebra null
python -m experiments.phoneme_concept_study --n-seeds 3

# NEGATIVE — pure base-10 emergence fails
python -m experiments.emergent_base10_study \
    --scan 50 100 --n-seeds 3

# CAUSAL — post-hoc bundle swap (Appendix B, ~1 min)
python -m experiments.counterfactual_swap_study --n-seeds 3
```

Total compute: < 25 min on a single RTX 4090 for the full four-domain replication (numbers + colors + space + phonemes + base-10 null + swap). All seeds and hyperparameters are shipped as JSON next to checkpoints.

12 Figures

All six auto-generated figures live in `mind/docs/research/figures/` (PDF + PNG). Regenerate in ~3 min with:

```
python -m experiments.render_paper_figures
```

#	File	What
F1	`F1_pcm_architecture.*` (diagram, to be drawn by hand)	PCM architecture: `muscle.forward(cid) → node.collapse(caller, facet) → ParamBundle`, with consumer registry and collapse history
F2	`F2_number_cos_heatmaps.*`	Number bundle 7×7 cos heatmaps — arithmetic_bias + ordinal_offset on a dual-muscle run; cross-facet align annotated
F4	`F4_four_domain_universality.*`	**Four-domain universality panel**: linear number line cos heatmap · circular hue ring MDS (colored by true HSV) · 2-D spatial grid MDS Procrustes-aligned to GT · phoneme cos heatmap with manner-class block lines
F5	`F5_base10_spike_null.*`	Base-10 emergence null: avg cos(n, n+k) vs k for N ∈ {50, 100} — smooth monotone line, no peak at k = 10
F6	`F6_swap_dissociation.*`	Counterfactual swap double-dissociation bars: 4 conditions × (inv / not-inv) × (muscle A / muscle B), both domains

#	File	What
F7	`F7_h5pp_alignment_schema.*`	**H5" four-domain schema**: six cross-facet-alignment bars (number arith ↔ ord · color mix ↔ adj · space motion ↔ L1 · phoneme v ↔ m / v ↔ p / m ↔ p) with permutation-null band and algebra-regime shading
F8	`F8_space_mds_trained_vs_shuffle.*`	Space grid MDS overlay, trained vs shuffle counterfactual: Procrustes disparity 0.07 → 0.97

13 Citations

- Barsalou L. W. 1999 · Perceptual Symbol Systems. BBS.
- Bricken T. et al. 2023 · Towards Monosemanticity: Decomposing Language Models with Dictionary Learning. Anthropic tech report.
- Caruana R. 1997 · Multitask Learning. Machine Learning 28.
- Chen Y. et al. 2022 · HyperPrompt: Prompt-based Task-Conditioning of Transformers. ICML.
- Dehaene S. 2011 · The Number Sense: How the Mind Creates Mathematics (revised ed.). Oxford University Press.
- Elhage N. et al. 2022 · Toy Models of Superposition. Transformer Circuits.
- Feigenson L., Dehaene S., Spelke E. 2004 · Core Systems of Number. Trends in Cognitive Sciences 8(7).
- Graves A. et al. 2016 · Hybrid Computing Using a Neural Network with Dynamic External Memory. Nature 538 (DNC).
- Ha D., Dai A., Le Q. 2017 · HyperNetworks. ICLR.
- Jacot A., Gabriel F., Hongler C. 2018 · Neural Tangent Kernel: Convergence and Generalization in Neural Networks. NeurIPS.
- Kaiser Ł., Sutskever I. 2016 · Neural GPUs Learn Algorithms. ICLR.
- Kazemnejad A. et al. 2023 · The Impact of Positional Encoding on Length Generalization in Transformers. NeurIPS.
- Khandelwal U., Levy O., Jurafsky D., Zettlemoyer L., Lewis M. 2020 · Generalization through Memorization: Nearest-Neighbor Language Models. ICLR (kNN-LM).
- Koh P. W. et al. 2020 · Concept Bottleneck Models. ICML.
- Madsen A., Johansen A. R. 2020 · Neural Arithmetic Units. ICLR.
- Maennel H. et al. 2020 · What Do Neural Networks Learn When Trained With Random Labels? NeurIPS.
- Marks L. et al. 2025 · Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models. ICLR.
- Maurer A. 2016 · The Benefit of Multitask Representation Learning. JMLR.

- McLeish S. et al. 2024 · Transformers Can Do Arithmetic with the Right Embeddings (Abacus). NeurIPS.
- von Oswald J. et al. 2020 · Continual Learning with Hypernetworks. ICLR.
- Pritzel A. et al. 2017 · Neural Episodic Control. ICML.
- Raghu M., Gilmer J., Yosinski J., Sohl-Dickstein J. 2017 · SVCCA: Singular Vector Canonical Correlation Analysis for Deep Learning Dynamics and Interpretability. NeurIPS.
- Ruder S. 2017 · An Overview of Multi-Task Learning in Deep Neural Networks. arXiv:1706.05098.
- Schmidhuber J. 1992 · Learning to Control Fast-Weight Memories. Neural Computation 4(1).
- Shallice T. 1988 · From Neuropsychology to Mental Structure. Cambridge University Press (double dissociation methodology).
- Templeton A. et al. 2024 · Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet. Anthropic tech report.
- Welch B. L. 1947 · The Generalisation of "Student's" Problem when Several Different Population Variances are Involved. Biometrika.
- Wittgenstein L. 1953 · Philosophical Investigations §43.
- Zarlenga M. E. et al. 2022 · Concept Embedding Models. NeurIPS.

14 Appendix A — Full raw numbers per study

(See `mind/docs/research/SINGLE_VS_DUAL_MUSCLE_FINDING.md`, `SCALE_STUDY.md`, `PURITY_AUDIT.md`, `QUAD_STUDY.md`, `COMPOSITIONAL_NUMBER_STUDY.md`, `COLOR_CONCEPT_STUDY.md`, `SPACE_CONCEPT_STUDY.md`, `PHONEME_CONCEPT_STUDY.md`, `EMERGENT_BASE10_STUDY.md`, `COUNTERFACTUAL_SWAP.md` for per-seed rows and analysis. All data are in `outputs/**/summary.json`.)

15 Appendix B — Counterfactual bundle swap (causal test)

All prior PCM evidence is either architectural-by-construction (attribution via `consumed_by`) or correlational / necessity (shuffle collapses ρ, permutation tests give low p). The strongest version of the PCM claim — **the bundle is the concept's semantic identity itself, not a learned correlate of it** — requires a post-hoc causal intervention.

Design (both domains): train dual-muscle models to ceiling, then swap the `.data` of two trained concepts' bundle parameters **in place**, at a specific facet only, and re-evaluate on the full pair set. This leaves head weights and optimizer state untouched; only the identity map (concept_id ↔ bundle tensor) is altered. Three interventions per seed: swap facet A only, swap facet B only, swap both. Three random seeds per domain.

B.1 Numbers (swap `concept:ans:3` ↔ `concept:ans:5`). Muscles: AddHead (`arithmetic_bias`, 64d) and CmpHead (`ordinal_offset`, 8d).

Condition	Add-inv	Add-not	Cmp-inv	Cmp-not
Baseline	100.0 %	100.0 %	100.0 %	100.0 %
Swap `arith_bias`	**18.2 %**	100.0 %	100.0 %	100.0 %
Swap `ord_offset`	100.0 %	100.0 %	**75.0 %**	100.0 %
Swap both	18.2 %	100.0 %	75.0 %	100.0 %

B.2 Colors (swap `concept:color:2` ↔ `concept:color:5`). Muscles: MixHead (`mixing_bias`, 64d) and AdjHead (`adjacency_offset`, 8d).

Condition	Mix-inv	Mix-not	Adj-inv	Adj-not
Baseline	100.0 %	100.0 %	100.0 %	100.0 %
Swap `mixing_bias`	**5.3 %**	100.0 %	100.0 %	100.0 %
Swap `adj_offset`	100.0 %	100.0 %	**23.8 %**	100.0 %
Swap both	5.3 %	100.0 %	23.8 %	100.0 %

All numbers are identical across 3 seeds ($\sigma = 0$) in both domains — full-pair enumeration at ceiling accuracy makes the post-swap prediction deterministic given the fixed pair-arithmetic.

Interpretation. The pattern is a textbook **double dissociation** at the parameter level:

1. Targeted collapse. A single-facet swap collapses accuracy only on pairs involving the swapped concepts, and only for the muscle that consumes that facet. The non-involved concepts stay at 100 % — the bundle tensor is fully "portable", it brings its semantic role with it.
2. Non-contamination. Swapping `arithmetic_bias` leaves CmpHead at 100 %; swapping `ordinal_offset` leaves AddHead at 100 %. The two facets carry cleanly separable shares of concept identity.
3. Non-random degradation. 18.2 % is not chance (chance ≈ 14.3 %); it is the residual correct rate expected when every "3" bundle is deterministically behaving as "5" and vice versa. The AddHead is computing the **swapped semantics**, not noise.
4. Domain-general. Numbers (linear algebra) and colors (circular algebra) produce the same qualitative pattern, ruling out linearity-specific explanations.

What this rules out. (a) Bundles are initialisation noise that training routes around — falsified, swap causes precisely-targeted drops. (b) Concept identity lives in head / backbone weights — falsified, heads are untouched and the targeted pairs collapse. (c) Facets leak — falsified, each facet-swap is invisible to the other muscle.

This is the cleanest causal evidence for PCM's central claim and closes the loop from correlational geometry (§4–§6) to mechanistic identity. Full numbers and discussion: `COUNTERFACTUAL_SWAP.md`. Code: `counterfactual_swap_study.py` (~1 min, 3 seeds, single GPU).